

Full trace evolution: Base -> Cand.-CE -> CE+SFT reply 0.15

20 selected cases from the final covered-only n=1,000 evaluation

- Objective: show how the complete generated trace (reasoning -> belief -> action) changes across training stages, tied to the paper claims.
- Latest arm: sft_r15_gx; warm-start from theta_ce; rho=0.15; covered-only evaluation completed at n=1,000.
- Valid three-arm, well-formed intersection: n=923.
- Selection mixes trajectory-continuation and non-continuation cases (9/11).
- Positive, regression, and persistent-failure cases are all included.

Extraction and selection protocol

1. Base: EGO_jihun3/runs/cesft_v2_fp/eval/base.records.jsonl
2. Cand.-CE: EGO_jihun3/runs/cesft_v2_fp/eval/theta_ce.records.jsonl
3. CE+SFT reply-0.15: EGO_jihun3/runs/cesft_v2_fp_gx/eval/sft_r15_gx.records.jsonl
4. Context: EGO_jihun3/runs/cesft_v2_fp/data/context_val.jsonl
5. Inner-join by sample_id; exclude any sample malformed in any arm.
6. Stratify by correctness pattern, G1/G2, and trajectory continuation.
7. Select across six qualitative buckets with continuation/non-continuation balance.

Selection mix

- G2 correction retained: 5
- G1 recovery retained: 3
- History-continuation success: 4
- SFT recovery beyond Cand.-CE: 3
- SFT regression from Cand.-CE: 3
- Persistent failure / boundary limit: 2

What to inspect

- Boundary discrimination versus generic plausibility.
- G1 retention and G2 correction.
- Use of completed-action history.
- Belief content beyond action echo.
- Reply-0.15 preservation, recovery, or regression.
- Failure modes delimiting the claim.

Final sft_r15_gx aggregate metrics

- Strict SelAcc: 28.8%
 - Conditional G1 / G2: 42.5% / 25.6%
 - Continuation recall / precision: 48.4% / 68.2%
 - Evidence mention / utility: 17.5% / +18.3pp
 - Belief-action echo: 0.0%
 - Paired SelAcc vs Cand.-CE: -0.64pp, 95% CI [-3.88,+2.91]
 - Paired GADR vs Cand.-CE: -2.53pp, 95% CI [-6.98,+2.29]
 - Non-harm gate: FAIL
 - Freegen in-support / GT-correct: 44.4% / 41.4% (n=500)
- GX-specific history-strip and belief-swap evaluations are unavailable; predecessor values are not attributed to GX.

Correctness patterns in valid three-arm traces

- 000 476 all wrong
- 001 98 SFT-only recovery
- 010 80 CE-only success; SFT regression
- 011 69 CE correction retained
- 100 48 Base-only success
- 101 23 Base+SFT correct
- 110 46 SFT regression from Base+CE
- 111 83 all correct

Case 01: G2 correction retained

Sample ID: 04fe8f4d-081e-437e-a56a-2d53b6233fc9_206
GT: add rice | WM top-1: stir ingredient | G2 (WM top-1 wrong) | GT rank: 8
Correctness [Base, CE, SFT]: 011 | trajectory continuation: False
Scenario: COOKING:MAKE_STIR_FRIED_DISH_GENERAL

Recent completed actions: add rice -> add rice -> add rice -> add rice -> add rice -> check heat

Shuffled Top-K: check heat, add rice, add spice_(seasoning), stir ingredient, add ingredient, turn stovetop, flip ingredient, mix_(beat,_blend,_whisk) ingredient, stir dough, add oil

WHAT TO INSPECT: Inspect how Cand.-CE rejects an incorrect WM leading proposal and whether CE+SFT preserves the corrected action using visual or trajectory evidence.

Base - WRONG

ACTION: add ingredient

BELIEF: Transferring the cooked item to a bowl.

REASONING: I am currently holding a cooked item in a ladle and moving it towards a metal bowl, likely to transfer it from the cooking pan. The previous action was checking the heat, and I have already added rice multiple times. The most logical next step is to transfer the cooked item to the bowl, which aligns with the action of 'add ingredient' as I am adding the cooked item to the bowl.

Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: add rice

BELIEF: Continue adding rice to the cooking pot.

REASONING: The current action involves checking the heat of the stove, which is a preparatory step before adding more ingredients. The previous actions have been focused on adding rice, indicating that the cooking process is ongoing. The next logical step would be to continue with the cooking process, likely by adding more rice or adjusting the heat. However, since the heat has just been checked, it is reasonable to assume that the next step would be to add more rice to continue the cooking process.

Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: add rice

BELIEF: I am in the middle of cleaning the pan after transferring the cooked item.

REASONING: I have just finished transferring the cooked item from the pan to a metal bowl, and the pan is now empty except for some residue. The bowl I just used is still in my hand, and I can see the cooked item inside it. The pan is on the stove, and I am holding the bowl with the cooked item. I need to clean the pan before adding more ingredients. The most logical next step is to clean the pan, which involves wiping it with a cloth or sponge.

Tags: trajectory wording=True | belief-action echo=False

Case 02: G2 correction retained

Sample ID: 1558e9f2-d7f3-4a23-9627-4240f506d7df_573
GT: roll dough | WM top-1: add topping | G2 (WM top-1 wrong) | GT rank: 2
Correctness [Base, CE, SFT]: 011 | trajectory continuation: False
Scenario: COOKING:MAKE_DUMPLINGS

Recent completed actions: flip ingredient -> add ingredient -> roll dough -> knead dough -> roll dough -> flip ingredient
Shuffled Top-K: add topping, roll dough, add ingredient, make dough, knead dough, check heat, sprinkle flour, put_(place,_set) dough, cut dough, shape dough
WHAT TO INSPECT: Inspect how Cand.-CE rejects an incorrect WM leading proposal and whether CE+SFT preserves the corrected action using visual or trajectory evidence.

Base - WRONG

ACTION: shape dough
BELIEF: Shape the flattened dough into the desired form.
REASONING: I have been repeatedly rolling and flipping the dough, and just added an ingredient. The dough is now spread out on the table, and I am currently pressing down on it with my hands, likely to flatten it further or prepare it for the next step. Given the context of preparing dough for a dish, the most logical next step is to shape it, as rolling has already been done multiple times and the dough is ready for forming.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: roll dough
BELIEF: Roll the dough to prepare it for the next step.
REASONING: I have been repeatedly rolling and flipping the dough, and adding ingredients. The current state shows the dough is on the table, and I've just flipped the ingredient. The most logical next step is to continue rolling the dough to achieve the desired thickness or texture.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: roll dough
BELIEF: I am in the middle of preparing the dough by flattening it further.
REASONING: I have been repeatedly rolling and flipping the dough, and the current state shows it is still on the table with no visible topping or additional ingredients added yet. The rolling pin is nearby, and the dough appears to need further flattening. Among the candidates, rolling dough is the only action that directly continues the established pattern of preparing the dough for the next step, which likely involves adding toppings or shaping. The other options, like adding ingredients or checking heat, are not supported by the current visual context or the sequence of actions.
Tags: trajectory wording=True | belief-action echo=False

Case 03: G2 correction retained

Sample ID: 478d7fa4-b174-4266-b0ff-bd180ba0b806_1650
GT: cook flatbread | WM top-1: transfer flatbread | G2 (WM top-1 wrong) | GT rank: 6
Correctness [Base, CE, SFT]: 011 | trajectory continuation: False
Scenario: COOKING:COOKING_GENERAL

Recent completed actions: check heat -> flip ingredient -> check heat -> flip ingredient -> roll dough -> check heat
Shuffled Top-K: transfer flatbread, sprinkle flour, add oil, add ingredient, knead dough, check heat, shape dough, cook flatbread, flip flatbread, cut dough
WHAT TO INSPECT: Inspect how Cand.-CE rejects an incorrect WM leading proposal and whether CE+SFT preserves the corrected action using visual or trajectory evidence.

Base - WRONG

ACTION: check heat
BELIEF: Monitor the heat to ensure proper cooking conditions for the flatbread.
REASONING: I am currently cooking a flatbread on the stove, and the previous action was checking the heat. The flatbread appears to be cooking, and I need to monitor its progress. The most logical next step is to check the heat again to ensure the cooking conditions are optimal before proceeding with any further actions like flipping or transferring.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: cook flatbread
BELIEF: Cook the flatbread on the stove.
REASONING: The current action history shows a repetitive cycle of rolling dough, checking heat, and cooking flatbread. The most recent action was checking heat, which suggests the next logical step is to continue cooking the flatbread. The flatbread is already on the stove, and the person is likely preparing to flip it or continue cooking it. Given the context, the next action should be to cook the flatbread further.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: cook flatbread
BELIEF: I am in the middle of cooking a flatbread on the stove.
REASONING: I am holding a piece of dough and have just finished rolling it, with the cooked flatbread visible on the stove. The fire is active and the stove is hot, indicating it's ready for the next item. Since I have a prepared dough ready to cook and the stove is hot, the most direct next step is to place it on the stove to cook.
Tags: trajectory wording=True | belief-action echo=False

Case 04: G2 correction retained

Sample ID: 0e6fb738-05fc-4dd5-9746-a8e10efe8c20_439
GT: check heat | WM top-1: stir dough | G2 (WM top-1 wrong) | GT rank: 6
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:MAKE_RICE_DISH_GENERAL

Recent completed actions: mix_(beat,_blend,_whisk) ingredient -> add ingredient -> add ingredient -> mix_(beat,_blend,_whisk) ingredient -> check heat -> check heat
Shuffled Top-K: check heat, add water, stir ingredient, clean surface, stir dough, knead dough, add oil, sieve ingredient, make dough, add ingredient
WHAT TO INSPECT: Inspect how Cand.-CE rejects an incorrect WM leading proposal and whether CE+SFT preserves the corrected action using visual or trajectory evidence.

Base - WRONG

ACTION: stir ingredient
BELIEF: Mixing the ingredients thoroughly to ensure even consistency.
REASONING: I am currently mixing ingredients in a bowl, and the previous actions indicate I have been preparing a dish that involves mixing and checking heat. The current frame shows me actively mixing, and the next logical step is to continue mixing to ensure the ingredients are well combined. Among the candidates, "stir ingredient" is the most appropriate as it matches the action I am performing.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: check heat
BELIEF: Monitor and adjust the stove heat for optimal cooking.
REASONING: The current action involves checking the heat of the stove, which is a preparatory step before cooking. The previous actions indicate that the cooking process is ongoing, and the next logical step is to ensure the heat is appropriate for the next stage of cooking. The candidate "check heat" aligns with this need to monitor and adjust the cooking conditions.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: check heat
BELIEF: I am in the middle of preparing the water for cooking by adjusting the stove heat.
REASONING: I have just finished mixing the yellowish mixture in the bowl and am now reaching for the pot of water on the stove. The water is already in the pot, and I am about to interact with it, likely to adjust the heat or prepare for the next step. The other bowls and ingredients are set aside, indicating I am focused on this specific pot. The action of checking the heat is the most direct and contextually appropriate next step given the current setup and my recent activity.
Tags: trajectory wording=True | belief-action echo=False

Case 05: G2 correction retained

Sample ID: 13c76616-f168-4af0-8d2a-fe82ce232d6a_527
GT: cook rice | WM top-1: add rice | G2 (WM top-1 wrong) | GT rank: 10
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:MAKE_STIR_FRIED_RICE

Recent completed actions: check heat -> cook rice -> check heat -> cook rice -> remove dirt -> cook rice
Shuffled Top-K: add rice, wash ingredient, check heat, wipe dish, stir dish, add salt, sieve ingredient, cook rice, turn stovetop, add water
WHAT TO INSPECT: Inspect how Cand.-CE rejects an incorrect WM leading proposal and whether CE+SFT preserves the corrected action using visual or trajectory evidence.

Base - WRONG

ACTION: stir dish
BELIEF: Continue cooking the rice by stirring it in the pot.
REASONING: I am currently stirring the rice in the pot with a wooden spoon, which is part of the cooking process. The previous actions show a repetitive pattern of cooking rice, checking heat, and removing dirt, indicating that the rice is being prepared for consumption. Since I am actively stirring the rice, the most logical next step is to continue cooking it, which aligns with the ongoing task of preparing the rice. The other options, such as adding water or salt, are not immediately necessary at this stage, as the rice is already in the pot and being stirred.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: cook rice
BELIEF: Continue cooking the rice.
REASONING: The current action history shows a repetitive cycle of cooking rice, checking heat, and removing dirt. The most recent action was removing dirt, which suggests the rice is being prepared for further cooking. The next logical step in this process is to continue cooking the rice, as the heat has been checked and the rice is already in the pot. The candidate 'cook rice' aligns with this ongoing task.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: cook rice
BELIEF: I am in the middle of cooking the rice in the pot.
REASONING: I have been repeatedly checking the heat and cooking the rice, and just removed dirt from the pot. The rice is still in the pot and I am holding the wooden paddle, indicating I am in the middle of stirring or adjusting the rice. The most consistent continuation of this activity is to keep cooking the rice, as the other options like adding water or salt are not supported by the current state of the pot or my tools.
Tags: trajectory wording=True | belief-action echo=False

Case 06: G1 recovery retained

Sample ID: 3bd5bf35-d6ac-43b2-ab75-1558a37c8550_1209
GT: fill bag | WM top-1: fill bag | G1 (WM top-1 correct) | GT rank: 1
Correctness [Base, CE, SFT]: 011 | trajectory continuation: False
Scenario: COOKING:MAKE_PASTRY

Recent completed actions: squeeze bag -> stir bread -> fill bag -> squeeze bag -> fill bag -> squeeze bag
Shuffled Top-K: stir bread, wash hand, fill bag, make filling, add ingredient, add salt, squeeze bag, strain liquid, knead dough, add grain
WHAT TO INSPECT: The WM top-1 is correct but Base misses it. Inspect whether candidate alignment learns to retain the valid prior and whether CE+SFT keeps that recovery.

Base - WRONG

ACTION: stir bread
BELIEF: Stirring the dough to ensure it is well-mixed and ready for the next step.
REASONING: I am currently stirring the dough in the plate, which is part of the ongoing process of preparing the mixture. The previous actions involved filling and squeezing bags, suggesting a repetitive task. Since I am actively mixing the dough, the next logical step is to continue this action to ensure the dough is well-prepared before proceeding to the next stage, which might involve filling or squeezing bags again. The candidate "stir bread" matches this context, as it implies continuing the mixing process.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: fill bag
BELIEF: Continue the cycle of filling and squeezing the bag.
REASONING: The current action is continuing the repetitive cycle of filling and squeezing the bag, which has been the pattern for the last several steps. The dough is being stirred, indicating preparation for another batch or continuation of the current task. The most logical next step is to continue the established pattern of filling the bag, as the previous actions suggest this is the ongoing subgoal.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: fill bag
BELIEF: I am in the middle of filling the bag with dough mixture.
REASONING: I have just finished squeezing the bag and am holding the plate with the dough mixture. The dough is still in the plate, and my hands are covered in it, indicating I'm not yet done with this step. The bag is empty and ready to be refilled, and the dough is still in the plate, suggesting I need to continue the process of filling and squeezing. The most logical next step is to fill the bag again with the dough mixture.
Tags: trajectory wording=True | belief-action echo=False

Case 07: G1 recovery retained

Sample ID: 3bd5bf35-d6ac-43b2-ab75-1558a37c8550_1260
GT: fill bag | WM top-1: fill bag | G1 (WM top-1 correct) | GT rank: 1
Correctness [Base, CE, SFT]: 011 | trajectory continuation: False
Scenario: COOKING:MAKE_PASTRY

Recent completed actions: fill bag -> squeeze bag -> fill bag -> squeeze bag -> fill bag -> squeeze bag
Shuffled Top-K: add paste, check heat, add ingredient, mix_(beat,_blend,_whisk) ingredient, make filling, stir bread, add grain, add salt, knead dough, fill bag
WHAT TO INSPECT: The WM top-1 is correct but Base misses it. Inspect whether candidate alignment learns to retain the valid prior and whether CE+SFT keeps that recovery.

Base - WRONG

ACTION: add paste
BELIEF: Mixing the paste in the bowl.
REASONING: The person is currently holding a spoon and stirring a white paste in a bowl, which suggests they are mixing or preparing a substance. Given the history of actions involving filling and squeezing bags, and the current context of mixing, the most logical next step is to continue mixing the paste. Among the candidates, "add paste" is the most relevant as it aligns with the current activity of preparing a mixture.
Tags: trajectory wording=False | belief-action echo=False

Cand.-CE - CORRECT

ACTION: fill bag
BELIEF: Continue the process of filling and squeezing the bag.
REASONING: The person has been repeatedly filling and squeezing a bag, likely for a repetitive task like applying a substance. The current frame shows them holding a plate with a white substance, possibly dough or paste, and a spoon. Given the pattern of actions, the next logical step is to continue the process of preparing or applying the substance, which aligns with 'fill bag' as the next action.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: fill bag
BELIEF: I am in the middle of preparing the bag for the next step in the process.
REASONING: I have just finished squeezing the bag and am holding the spatula over the bowl, which still contains the white paste. The bowl is positioned near the bag, and my hands are ready to interact with it again. Among the candidates, only 'fill bag' directly follows the current state of having just squeezed the bag and having the paste accessible. The other actions like adding ingredients or mixing don't align with the immediate context of the bag and paste.
Tags: trajectory wording=True | belief-action echo=False

Case 08: G1 recovery retained

Sample ID: bb2297d3-91b7-4ec4-b39b-cc64694929db_3237
GT: organize_(arrange) tool_(utensil) | WM top-1: organize_(arrange) tool_(utensil) | G1 (WM top-1 correct) | GT rank: 1
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:COOKING_GENERAL

Recent completed actions: unpack ingredient -> organize_(arrange) tool_(utensil) -> organize_(arrange) tool_(utensil) -> organize_(arrange) tool_(utensil)
Shuffled Top-K: store ingredient, add water, turn stovetop, wash dish, add oil, add spice_(seasoning), get_(fetch,_take) ingredient, organize_(arrange) tool_(utensil), check heat, wipe dish

WHAT TO INSPECT: The WM top-1 is correct but Base misses it. Inspect whether candidate alignment learns to retain the valid prior and whether CE+SFT keeps that recovery.

Base - WRONG

ACTION: store ingredient
BELIEF: Store the ingredient I have been handling.
REASONING: I am currently in a bathroom, holding a bowl with what appears to be ingredients or tools, and I have just moved from the sink area toward the hallway. Given that I have already completed several organizing actions and I am now holding a bowl, it is likely I am preparing to store or use the contents. The most logical next step, considering the context and my current position, is to store the ingredient I have been handling.
Tags: trajectory wording=False | belief-action echo=False

Cand.-CE - CORRECT

ACTION: organize_(arrange) tool_(utensil)
BELIEF: Organizing tools and utensils in the kitchen.
REASONING: The current view shows a hallway with a dog, and the previous actions were all about organizing tools and unpacking ingredients. The context suggests we are in a kitchen or similar area, and the next logical step would be to continue organizing or preparing for cooking. Since 'organize_(arrange) tool_(utensil)' was the last action, repeating it makes sense to maintain consistency in the task.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: organize_(arrange) tool_(utensil)
BELIEF: I am in the middle of organizing tools and ingredients in the kitchen area.
REASONING: I have just finished arranging utensils and am now holding a container with what appears to be ingredients or tools. The camera pans to show a dog in the hallway, indicating I've moved away from the immediate workspace. The presence of a pink bucket and a white bowl on the floor suggests I'm in a transitional space, possibly moving between tasks. Given the context of organizing tools and the items I'm holding, the most logical next step is to continue arranging or storing these items.
Tags: trajectory wording=True | belief-action echo=False

Case 09: History-continuation success

Sample ID: 9adc0c59-1167-4311-abf3-7b57a0743b1d_2628
GT: bake ingredient | WM top-1: put_(place,_set) dough | G2 (WM top-1 wrong) | GT rank: 7
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:MAKE_BREAD

Recent completed actions: bake ingredient -> stir ingredient -> add water -> check ingredient -> add water -> bake ingredient
Shuffled Top-K: make dough, bake ingredient, knead dough, remove dough, roll dough, check heat, store ingredient, put_(place,_set) dough, add ingredient, organize_(arrange) ingredient
WHAT TO INSPECT: GT repeats the most recent completed action. Inspect explicit use of repetition/continuation rather than generic procedural common sense.

Base - WRONG

ACTION: check heat
BELIEF: Check the oven's heat before proceeding with the next baking step.
REASONING: The current view shows the person using a long tool to interact with something inside an oven or large appliance, likely checking or adjusting something. Given the history of baking actions and the context of the current frame, the most logical next step is to check the heat or status of the oven to ensure proper baking conditions. This aligns with the need to monitor the baking process after placing ingredients inside.
Tags: trajectory wording=False | belief-action echo=False

Cand.-CE - CORRECT

ACTION: bake ingredient
BELIEF: Continue the baking process by baking another ingredient.
REASONING: The current action history shows a sequence of baking-related tasks, including kneading, adding water, proofing, and baking ingredients. The most recent action was baking an ingredient, suggesting the process is ongoing. The environment appears to be a kitchen or bakery setting with equipment for baking. Given the context, the next logical step would be to continue the baking process or prepare for the next batch. Among the candidates, "bake ingredient" is the most consistent with the ongoing task and the recent action history.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: bake ingredient
BELIEF: I am in the middle of preparing the oven for baking by adjusting its interior.
REASONING: I have just finished placing the dough into the oven and am now using a long tool to adjust or check something inside. The motion suggests I'm ensuring the dough is properly positioned or verifying the oven's internal state. Given the context of baking and the tool's use, I'm likely confirming the oven's readiness or the dough's placement before closing it.
Tags: trajectory wording=True | belief-action echo=False

Case 10: History-continuation success

Sample ID: 9adc0c59-1167-4311-abf3-7b57a0743b1d_2641
GT: roll dough | WM top-1: knead dough | G2 (WM top-1 wrong) | GT rank: 2
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:MAKE_BREAD

Recent completed actions: knead dough -> roll dough -> sprinkle flour -> roll dough -> put_(place,_set) dough -> roll dough
Shuffled Top-K: knead dough, proof dough, organize_(arrange) ingredient, roll dough, cut dough, put_(place,_set) dough, make dough, squeeze bag, sprinkle flour, bake ingredient
WHAT TO INSPECT: GT repeats the most recent completed action. Inspect explicit use of repetition/continuation rather than generic procedural common sense.

Base - WRONG

ACTION: cut dough
BELIEF: Cutting the rolled dough into individual portions.
REASONING: I have just finished rolling dough and placing it on a surface. The current view shows me preparing to work with more dough, likely needing to shape or portion it. Given the sequence of actions, I am likely moving to the next step in preparing the dough for baking, which would involve cutting it into portions.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: roll dough
BELIEF: Rolling the dough to achieve the desired shape and thickness.
REASONING: The current view shows the dough being rolled out on a floured surface, indicating that the rolling process is ongoing. The previous action was 'roll dough', and the next logical step in the dough preparation sequence is to continue rolling or to prepare for the next stage, which could involve cutting or shaping. However, since the dough is still being rolled, continuing to roll is the most immediate and consistent action.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: roll dough
BELIEF: I am in the middle of preparing dough for shaping or further processing.
REASONING: I have just finished rolling dough and am now holding a piece of it, with the rolling surface and other dough pieces visible nearby. The dough appears to be at a stage where it needs to be shaped or prepared for the next step, and I am positioned to work with it directly. Among the candidates, rolling dough is the only one that matches the current state of having dough ready on the surface and my hands positioned to manipulate it. The other actions, like kneading or cutting, are not supported by the current visual context of the dough's state and my posture.
Tags: trajectory wording=True | belief-action echo=False

Case 11: History-continuation success

Sample ID: a8486c50-4576-41e3-a9ad-ee809e40c648_2700
GT: mix_(beat,_blend,_whisk) ingredient | WM top-1: wash hand | G2 (WM top-1 wrong) | GT rank: 3
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:MAKE_TACO

Recent completed actions: cut meat -> blanch ingredient -> cut meat -> mix_(beat,_blend,_whisk) ingredient
Shuffled Top-K: stir dish, dispose item, store ingredient, mix_(beat,_blend,_whisk) ingredient, wash tool_(utensil), make batter, wash hand, use phone, stir sauce, add oil

WHAT TO INSPECT: GT repeats the most recent completed action. Inspect explicit use of repetition/continuation rather than generic procedural common sense.

Base - WRONG

ACTION: wash hand
BELIEF: Clean hands after handling food
REASONING: I have just finished mixing the ingredients on the plate, and now I need to clean up. The most logical next step is to wash my hands since I've been handling food. The other options like stirring or adding oil don't make sense right now as I'm not actively cooking or preparing to cook.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: mix_(beat,_blend,_whisk) ingredient
BELIEF: Continue preparing the dish by mixing the ingredients.
REASONING: I have been preparing ingredients and mixing them. The current state shows I'm still working with the meat and other components on the plate. The most logical next step is to continue preparing the dish by mixing the ingredients together, which aligns with my previous actions and the current setup.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: mix_(beat,_blend,_whisk) ingredient
BELIEF: I am in the middle of seasoning the dish with a prepared ingredient.
REASONING: I have just finished handling the meat and am now reaching for the small white bowl containing a white substance, likely salt or seasoning, which is positioned near the other prepared ingredients. The bowl is within easy reach, and my hand movement suggests I am about to pick it up. Given the context of preparing ingredients and the presence of seasoning, the most logical next step is to use this ingredient, likely to season the meat or another component of the dish. The other candidates, like washing or storing, are not supported by the current setup or my hand position.
Tags: trajectory wording=True | belief-action echo=False

Case 12: History-continuation success

Sample ID: cc575a16-64fd-4cda-9248-5d85f506fdfd_3615
GT: grate ingredient | WM top-1: wash bowl | G2 (WM top-1 wrong) | GT rank: 4
Correctness [Base, CE, SFT]: 011 | trajectory continuation: True
Scenario: COOKING:COOKING_GENERAL

Recent completed actions: grate ingredient -> grate ingredient -> grate ingredient -> grate ingredient -> grate ingredient -> grate ingredient
Shuffled Top-K: sieve ingredient, wash ingredient, grate ingredient, wash bowl, add water, add spice_(seasoning), perform activity, cut ingredient, organize_(arrange) tool_(utensil), wash rice
WHAT TO INSPECT: GT repeats the most recent completed action. Inspect explicit use of repetition/continuation rather than generic procedural common sense.

Base - WRONG

ACTION: sieve ingredient
BELIEF: Sieve the grated ingredient to refine its texture.
REASONING: I have been repeatedly grating an ingredient, and the current state shows me holding a small bowl with grated material and a spoon. The next logical step in preparing this ingredient is to sieve it to separate finer particles from larger ones, which is a common step after grating. This aligns with the typical sequence of food preparation tasks.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: grate ingredient
BELIEF: Continue grating the ingredient as part of the ongoing preparation task.
REASONING: The current action history shows a repetitive pattern of grating an ingredient, suggesting the task is ongoing and likely nearing completion. The environment and setup remain consistent, indicating no significant change in context. The most logical next step, given the repetitive nature of the task, is to continue grating the ingredient until the desired amount is achieved. This aligns with the established pattern and the current state of the task.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: grate ingredient
BELIEF: I am in the middle of grating an ingredient into a bowl.
REASONING: I have been repeatedly grating an ingredient into a bowl, and the current state shows the grater and bowl still in use. The action of grating is ongoing, and the candidate actions include continuing to grate or switching to other tasks like sieving or washing. Since the grater is still in hand and the bowl contains grated material, the most consistent continuation of the current subtask is to keep grating. The other candidates, like sieving or washing, are not visually supported by the current setup or the state of the ingredients.
Tags: trajectory wording=True | belief-action echo=False

Case 13: SFT recovery beyond Cand.-CE

Sample ID: 0e0d6704-1c6c-4a62-bc97-cc55658cf8ac_355
GT: sprinkle flour | WM top-1: put_(place,_set) dough | G2 (WM top-1 wrong) | GT rank: 3
Correctness [Base, CE, SFT]: 001 | trajectory continuation: False
Scenario: COOKING:MAKE_BREAD

Recent completed actions: cut dough -> knead dough -> make dough -> cut dough -> knead dough -> organize_(arrange) sheet
Shuffled Top-K: knead dough, organize_(arrange) sheet, organize_(arrange) tool_(utensil), cut dough, sprinkle flour, put_(place,_set) dough, add ingredient, roll dough, check heat, shape dough

WHAT TO INSPECT: Cand.-CE is still wrong but CE+SFT becomes correct. Inspect whether the new belief organizes evidence that the CE trace overlooked.

Base - WRONG

ACTION: check heat
BELIEF: Check the oven’s heat before baking the dough.
REASONING: I have just finished organizing the sheet and am now looking at the work area. The dough is already prepared and placed on trays, and I need to continue with the next step in the baking process. Since I’ve been working with dough and the area is ready, the logical next step is to check the oven’s heat to ensure it’s properly preheated for baking.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - WRONG

ACTION: cut dough
BELIEF: Cutting the dough to prepare it for the next step.
REASONING: I have been repeatedly kneading, cutting, rolling, and placing dough, and I just finished organizing the sheet. The next logical step in this repetitive process is to continue preparing the dough by cutting it again, as this is part of the ongoing task of preparing dough for baking.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: sprinkle flour
BELIEF: I am preparing the work surface for shaping the dough.
REASONING: I have just finished arranging the dough on the sheet and am now looking at the work surface, which is covered in flour and has a few stray pieces of dough. The dough I just placed is still in its raw, unshaped state, and I need to prepare the surface for the next step. Sprinkling flour will prevent sticking and make it easier to handle the dough as I shape it.
Tags: trajectory wording=True | belief-action echo=False

Case 14: SFT recovery beyond Cand.-CE

Sample ID: 228eb02a-6c89-4aa0-9cd4-0cdab1550c83_820
GT: stir sauce | WM top-1: stir vegetable | G2 (WM top-1 wrong) | GT rank: 5
Correctness [Base, CE, SFT]: 001 | trajectory continuation: False
Scenario: COOKING:MAKE_RICE_DISH_GENERAL

Recent completed actions: add sauce-> check instruction -> stir sauce-> add pepper -> check heat -> add pepper
Shuffled Top-K: check heat, stir ingredient, stir dish, mix_(beat_blend_whisk) ingredient, cook dish, stir sauce, organize_(arrange) ingredient, stir onion, stir vegetable, boil noodle

WHAT TO INSPECT: Cand.-CE is still wrong but CE+SFT becomes correct. Inspect whether the new belief organizes evidence that the CE trace overlooked.

Base - WRONG

ACTION: stir dish
BELIEF: Stir the dish to mix ingredients and ensure even cooking.
REASONING: I have just finished adding pepper and checking the heat. The current state shows the pan with ingredients on the stove, and I am holding a wooden spoon, indicating I am ready to stir the contents. Among the candidates, "stir dish" is the most logical next step to continue cooking and ensure even heating and mixing of ingredients.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - WRONG

ACTION: stir dish
BELIEF: Stir the dish to mix ingredients and cook evenly.
REASONING: I have just added pepper to the dish and checked the heat. The current state shows the pan on the stove with ingredients inside, and I need to continue cooking. The most logical next step is to stir the dish to ensure even cooking and to incorporate the newly added pepper.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: stir sauce
BELIEF: I am in the middle of blending the sauce in the pan to evenly distribute the added pepper and other seasonings.
REASONING: I have just finished adding pepper and am holding the cutting board, which suggests I've completed the chopping phase. The pan on the stove contains a mixture of vegetables and sauce, and I'm looking at the ingredients on the counter, indicating I'm assessing what's next. Among the candidates, "stir sauce" is the most consistent with the current state of the dish, which appears to be simmering and needs blending to incorporate the added pepper and other seasonings. The other options like organizing or boiling noodles don't align with the current cooking context.
Tags: trajectory wording=True | belief-action echo=False

Case 15: SFT recovery beyond Cand.-CE

Sample ID: 2c0c6508-397f-4c48-aeb7-abc7a3cae8d1_1102
GT: cut garlic | WM top-1: stir ingredient | G2 (WM top-1 wrong) | GT rank: 3
Correctness [Base, CE, SFT]: 001 | trajectory continuation: True
Scenario: COOKING:MAKE_BURRITO

Recent completed actions: make filling -> wash pepper -> cut pepper -> add spice_(seasoning) -> stir chicken -> cut garlic
Shuffled Top-K: dispose item, cut ingredient, wash ingredient, peel garlic, add spice_(seasoning), cut garlic, mix_(beat,_blend,_whisk) ingredient, cut pepper, stir ingredient, wash hand

WHAT TO INSPECT: Cand.-CE is still wrong but CE+SFT becomes correct. Inspect whether the new belief organizes evidence that the CE trace overlooked.

Base - WRONG

ACTION: dispose item
BELIEF: Dispose of the garlic waste.
REASONING: I am currently holding a small piece of garlic that I have just cut. My hands are near the sink, and I am looking at the trash bin. The most logical next step is to dispose of the garlic peel or waste, as I have already cut the garlic and it is no longer needed for the recipe.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - WRONG

ACTION: peel garlic
BELIEF: peel garlic
REASONING: I have just cut garlic and am now holding it in my hand. The next logical step is to prepare the garlic for cooking, which typically involves peeling it. I see the garlic in my hand, and peeling it is the most immediate and relevant action to continue the cooking process.
Tags: trajectory wording=True | belief-action echo=True

CE+SFT reply 0.15 (GX) - CORRECT

ACTION: cut garlic
BELIEF: I am in the middle of preparing garlic for use in the dish.
REASONING: I have just finished cutting garlic and am holding the pieces in my hand. The knife and cutting board are still in front of me, and the garlic is visibly separated from the clove. Among the candidates, cutting garlic is the only one that matches the current state of having just separated the garlic and being positioned to handle it further. The other actions, like washing or disposing, don't align with the immediate visual context of handling the garlic.
Tags: trajectory wording=True | belief-action echo=False

Case 16: SFT regression from Cand.-CE

Sample ID: 1a327b0c-b78b-4ae2-add0-49334bccddcd_730
GT: add spice_(seasoning) | WM top-1: get_(fetch,_take) ingredient | G2 (WM top-1 wrong) | GT rank: 2
Correctness [Base, CE, SFT]: 010 | trajectory continuation: False
Scenario: COOKING:MAKE_STEW_OR_SOUP

Recent completed actions: stir ingredient -> serve food -> remove ingredient -> flip ingredient
Shuffled Top-K: organize_(arrange) ingredient, add butter, toast bread, get_(fetch,_take) ingredient, use phone, get_(fetch,_take) tool_(utensil), organize_(arrange) tool_(utensil), preheat pan, add water, add spice_(seasoning)
WHAT TO INSPECT: A required negative case: Cand.-CE is correct and CE+SFT is wrong. Inspect hallucinated procedure, over-elaboration, or belief drift; this illustrates the non-harm failure.

Base - WRONG

ACTION: get_(fetch,_take) ingredient
BELIEF: Retrieve an ingredient or tool from the cabinet to continue cooking.
REASONING: I am currently in a kitchen, looking into an open cabinet. My hand is reaching inside, suggesting I am retrieving something. Given the context of cooking and the items visible (bowl, carrots, foil), I am likely getting an ingredient or tool to continue preparing the meal. The cabinet contains jars, which are commonly used for spices or seasonings. The most logical next step is to retrieve an ingredient or tool from the cabinet to proceed with the cooking task.
Tags: trajectory wording=False | belief-action echo=False

Cand.-CE - CORRECT

ACTION: add spice_(seasoning)
BELIEF: Add seasoning to the food.
REASONING: I have just finished flipping an ingredient, and I am now reaching into a cabinet. The cabinet appears to contain spices or seasonings, which suggests I am likely retrieving an ingredient to add to my dish. Given the context of cooking and the items visible on the counter, adding a spice is a logical next step to enhance the flavor of the food I am preparing.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - WRONG

ACTION: get_(fetch,_take) ingredient
BELIEF: I am in the middle of retrieving an ingredient from the cabinet.
REASONING: I have just finished flipping the ingredient and am now reaching into the cabinet, which contains jars and containers. The cabinet is open and I am extending my hand toward the items inside, suggesting I am retrieving something. The other candidates like organizing or using a tool don't align with the current motion of reaching into the cabinet. The most direct interpretation is that I am fetching an ingredient from the cabinet.
Tags: trajectory wording=True | belief-action echo=False

Case 17: SFT regression from Cand.-CE

Sample ID: 2a2ff7db-5460-4296-a8a7-946ba628226d_1016
GT: roll dough | WM top-1: put_(place,_set) dough | G2 (WM top-1 wrong) | GT rank: 6
Correctness [Base, CE, SFT]: 110 | trajectory continuation: False
Scenario: COOKING:MAKE_PASTRY

Recent completed actions: knead dough -> knead dough -> cut dough -> roll dough -> wash hand -> knead dough
Shuffled Top-K: add fruit, wrap leaf, put_(place,_set) dough, dough it, grate ingredient, cut mango, knead dough, put_(place,_set) amount, add ingredient, roll dough
WHAT TO INSPECT: A required negative case: Cand.-CE is correct and CE+SFT is wrong. Inspect hallucinated procedure, over-elaboration, or belief drift; this illustrates the non-harm failure.

Base - CORRECT

ACTION: roll dough
BELIEF: Shape the dough into the desired form.
REASONING: I am currently holding a small piece of dough and shaping it with my hands. The previous actions involved kneading and cutting dough, and I just finished washing my hands. The most logical next step is to continue working with the dough, likely by rolling it into a specific shape or size. Among the candidates, "roll dough" directly matches the current activity and the context of preparing dough.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: roll dough
BELIEF: Shape the dough into the desired form.
REASONING: I have been kneading dough and have just finished cutting and rolling some. The current frame shows me holding a piece of dough, likely preparing to shape or use it. Given the context of ongoing dough preparation, the most logical next step is to continue shaping or preparing the dough, which aligns with 'roll dough'.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - WRONG

ACTION: put_(place,_set) dough
BELIEF: I am in the middle of shaping a small piece of dough into a specific form.
REASONING: I have just finished kneading the dough and am now holding a small piece, preparing to shape it. The dough is still soft and pliable, and I'm focusing on forming it into a specific shape. The other candidates like cutting or rolling don't align with the current state of having a small, shaped piece ready. Putting it down would be the logical next step to set it aside before proceeding to the next item.
Tags: trajectory wording=True | belief-action echo=False

Case 18: SFT regression from Cand.-CE

Sample ID: 16d55886-6e1e-4195-9918-12dc4568320e_651
GT: use phone | WM top-1: perform activity | G2 (WM top-1 wrong) | GT rank: 3
Correctness [Base, CE, SFT]: 010 | trajectory continuation: True
Scenario: COOKING:COOKING_GENERAL

Recent completed actions: add ingredient -> add ingredient -> add salt -> add water -> use phone -> use phone
Shuffled Top-K: clean surface, store ingredient, organize_(arrange) tool_(utensil), clean area, add water, clean floor, use phone, wash tool_(utensil), perform activity, wash bowl
WHAT TO INSPECT: A required negative case: Cand.-CE is correct and CE+SFT is wrong. Inspect hallucinated procedure, over-elaboration, or belief drift; this illustrates the non-harm failure.

Base - WRONG

ACTION: clean area
BELIEF: Clean the workspace after using the phone.
REASONING: I have just finished using my phone, and the current view shows a plain surface with no immediate task or object requiring attention. The most logical next step is to clean the area to maintain hygiene or prepare for the next task. Among the candidates, "clean area" is the most appropriate as it directly addresses the need to tidy up the workspace after using the phone.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - CORRECT

ACTION: use phone
BELIEF: Continue using the phone after recent usage.
REASONING: The user has been repeatedly adding ingredients and water, and has just used their phone twice. The current frame shows the user holding a phone, suggesting they are still engaged with it. Given the pattern of recent actions, using the phone again is the most consistent next step.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - WRONG

ACTION: clean area
BELIEF: I am in the middle of tidying up the workspace after completing a cooking task.
REASONING: I have just finished using my phone and am now holding it in my hand, with the screen still visible. The surface I'm standing near appears to have some residue or debris, and I'm looking down at it. Among the candidates, cleaning the area aligns with the current state of the surface and the recent activity of using a phone, which might have left marks or smudges. The other options like organizing tools or washing a bowl don't match the immediate visual context of the surface needing attention.
Tags: trajectory wording=True | belief-action echo=False

Case 19: Persistent failure / boundary limit

Sample ID: 0a01978c-e16d-4587-95f1-49efa3ab15d9_284
GT: serve food | WM top-1: serve dish | G2 (WM top-1 wrong) | GT rank: 7
Correctness [Base, CE, SFT]: 000 | trajectory continuation: False
Scenario: COOKING:MAKE_PANCAKE

Recent completed actions: flip pancake -> flip pancake -> strain liquid -> pour batter -> check doneness -> stir bread
Shuffled Top-K: fry egg, organize_(arrange) tool_(utensil), stir ingredient, clean surface, serve dish, wash tool_(utensil), serve meat, serve food, mix_(beat, blend, whisk) ingredient, pour batter

WHAT TO INSPECT: All three arms fail despite GT being in the Top-K boundary. Inspect ambiguous evidence, misleading history, or a candidate distinction the language policy never resolves.

Base - WRONG

ACTION: organize_(arrange) tool_(utensil)
BELIEF: Organizing the cooking tools after finishing the pancake cooking process.
REASONING: I have been flipping pancakes and checking their doneness, indicating I am in the process of cooking breakfast. The current frame shows me finishing a flip and moving away from the stove, likely to assess the pancake or prepare for the next step. The most logical next action is to organize the tools I've used, as I've just finished cooking and need to clean up before serving. This aligns with typical kitchen workflow after cooking.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - WRONG

ACTION: pour batter
BELIEF: Prepare the next pancake by pouring batter.
REASONING: I have been repeatedly flipping pancakes and checking their doneness, indicating I am in the process of cooking breakfast. The current frame shows me just finished checking the doneness of the pancake. The most logical next step is to continue cooking or preparing the next item, which would be to pour more batter for another pancake, as I have already prepared the batter and the pan is ready.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - WRONG

ACTION: stir ingredient
BELIEF: I am in the middle of cooking pancakes and preparing to manage the batter for the next one.
REASONING: I have just finished checking the doneness of the pancake and am now holding the spatula over the pan. The batter container and spoon are visible on the counter nearby, and I have been repeatedly flipping the pancake, indicating I am in the middle of cooking it. The most logical next step, given the tools and the ongoing cooking task, is to continue managing the cooking process by stirring the batter in the container to prepare for the next pancake or to adjust consistency.
Tags: trajectory wording=True | belief-action echo=False

Case 20: Persistent failure / boundary limit

Sample ID: 106e9447-8a09-4e89-964e-61c927827a56_463
GT: dispose item | WM top-1: organize_(arrange) tool_(utensil) | G2 (WM top-1 wrong) | GT rank: 6
Correctness [Base, CE, SFT]: 000 | trajectory continuation: False
Scenario: COOKING:MAKING_MULTIPLE_RECIPES

Recent completed actions: store ingredient -> check doneness -> get_(fetch,_take) ingredient -> add ingredient -> serve drink -> get_(fetch,_take) tool_(utensil)
Shuffled Top-K: get_(fetch,_take) ingredient, check heat, grate ingredient, organize_(arrange) tool_(utensil), dispose item, wipe dish, wash hand, measure ingredient, organize_(arrange) ingredient, store ingredient

WHAT TO INSPECT: All three arms fail despite GT being in the Top-K boundary. Inspect ambiguous evidence, misleading history, or a candidate distinction the language policy never resolves.

Base - WRONG

ACTION: wipe dish
BELIEF: Clean the dish after arranging the ingredients.
REASONING: I have just finished arranging the sliced tomatoes in the dish, and now I need to clean up the workspace. The most logical next step is to wipe the dish to remove any stray tomato pieces or residue, ensuring the area is tidy before moving on to the next task.
Tags: trajectory wording=True | belief-action echo=False

Cand.-CE - WRONG

ACTION: organize_(arrange) tool_(utensil)
BELIEF: Organize the tools used in the cooking process.
REASONING: I have just finished adding an ingredient to the dish and am now looking at the tools on the counter. The next logical step in my cooking process is to organize or clean up the tools I've used, which aligns with the action of organizing utensils. This will help keep the workspace tidy and prepare for the next steps.
Tags: trajectory wording=True | belief-action echo=False

CE+SFT reply 0.15 (GX) - WRONG

ACTION: get_(fetch,_take) ingredient
BELIEF: I am in the middle of gathering and preparing ingredients for the meal.
REASONING: I have just finished arranging the sliced tomatoes in the dish and am now looking at the ingredients on the counter. The green bag of vegetables is still there, and I need to decide whether to use it next. Since I've already handled the tomatoes and the dish is ready, the most logical next step is to retrieve another ingredient to continue preparing the meal.
Tags: trajectory wording=True | belief-action echo=False